Impact of Tight Channel Spacing with Fiber Switching in WDM Networks using 800ZR+ Interfaces

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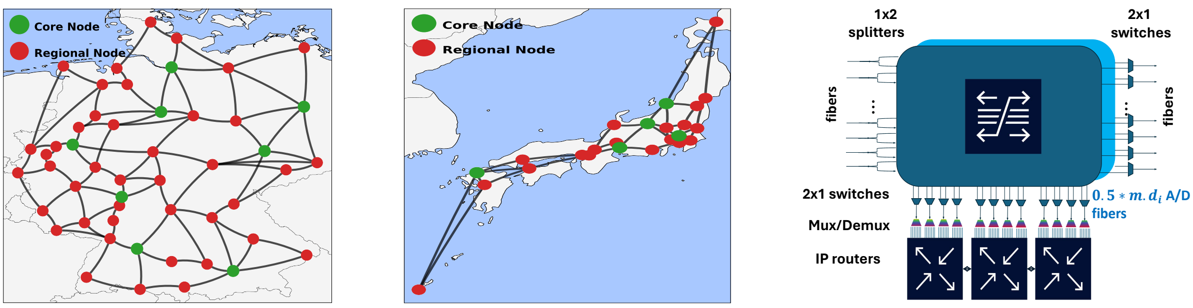
Abstract: This paper investigates network performance with pure fiber switching, exploring tighter channel spacings unconstrained by wavelength-routing granularity. It evaluates adjacent crosstalk and nonlinear transmission penalties using 800ZR+ coherent pluggable interfaces with two core topologies. © 2026 The Author(s)

1. Introduction

Pure fiber switching combined with a large bundle of fibers per link is one of the envisaged next-gen approaches to accommodate the unceasing growing demand of services in wavelength division multiplexd (WDM) networks. In this context, our prior work [1] compared the network performance of fiber vs. wavelength switching and explained the conditions under which the former can carry as much traffic as the latter, while allowing lower cost and lower physical impairments. This paper examines a further optimization allowed by pure fiber switching. It consists of tighter arbitrary channel spacing no longer constrained by the spectral granularity of wavelength-routing elements. Degradations to consider when reducing channel spacing are mainly the adjacent crosstalk (ADXT) incurred in the multiplexing stages of the source and destination nodes of the signals, as well as the WDM non-linear effects of transmission [2]. This study investigates their impact as a function of channel spacing on the global network performance for two different core WDM networks (German and Japanese topologies with 10 bidirectional fibers/link see Fig. 1a and Fig. 1b [3]) equipped with 800ZR+ coherent pluggable interfaces.

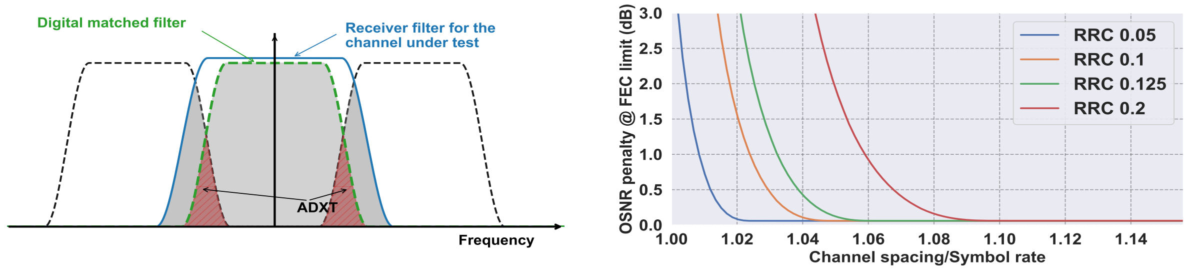
2. Network Architecture

The scalabilty of conventional wavelength switched networks with multiple parallel fibers per link is constrained by the limited number of WSS ports, typically capping the optical node degrees below 30. In contrast, pure fiber-switching architecture based on large non-blocking fiber cross-connects (FXCs) eliminates wavelength-level switching and can directly connect the IP layer to the fiber layer through optical mux/demux units (See Fig. 1c). This FXC-based network removes the WSS filtering penalty of reconfigurable optical add/drop multiplexers and supports over 1000×1000 fiber connections [4], enhancing scalability and network performance while reducing optical complexity. Fig. 2a illustrates the optical spectrum in the context of tightly spaced channels enabled by pure fiber switching. With pure fiber switching, channels can be packed closer in frequency, which increases the ADXT between neighboring carriers. This crosstalk originates from the leakage of adjacent channels into the bandwidth of the central received channel, as depicted by the blue region in Fig. 2a. In this work, we perform a network-level analysis to determine the conditions under which the ADXT degrades performance in terms of maximum network capacity (MNC). In this direction, Fig. 2b shows the optical signal-to-noise ratio (OSNR) penalty at the forward error correction (FEC) limit as a function of the normalized channel spacing (Δf/Symbol rate (S_r)) for various root-raised-cosine (RRC) roll-off factors of 800ZR+ coherent pluggable interfaces over the RRC range defined in [6]. Consistent with [7], narrower pulse shaping (RRC = 0.05 to 0.1) enables denser packing with minimal OSNR penalty down to about $\Delta f/S_r \approx 1.02$–$1.04$, while larger roll-off factors (≥ 0.125) result in



(a) (b) (c)

Fig. 1: *(a) German 50 nodes topology and (b) Japan 25 nodes topology, both with 10 bidirectional fibers/link (c) FXC architecture node equipped with ingress 1x2 splitters and egress 2x1 switches for FXC protection.*



(a) **(b)**

Fig. 2: *(a) The ADXT in signal spectrum at coherent receiver [5], (b) Simulated OSNR penalty vs. normalized channel spacing for 800ZR+ 800Gb/s channel @ 118 Gbaud.*

Grid [GHz]	150	145.5	141.18	137.14	133.33	129.73	126.31	123.07	120
# Channels	32	33	34	35	36	37	38	39	40

Data rate	800G/400G	800G/600G
Modulation	16QAM/QPSK	16QAM (PCS)
Symbol Rate (S_r) [Gbaud]	118.12	131.35/118.74

Table 1: *Grid channel spacing, number of WDM channels, and symbol rates for different modulation formats.*

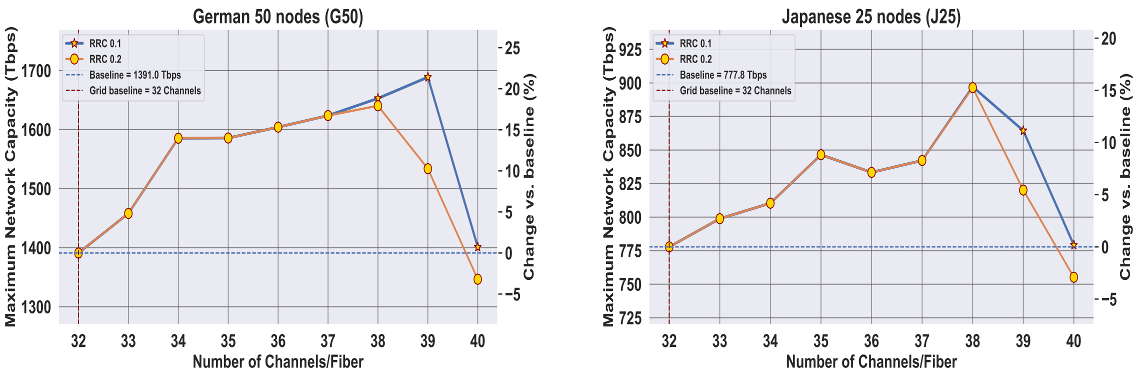
a smother penalty rise due to increased ADXT. In addition, reducing channel spacing enhances nonlinear effects such as self- and cross-phase modulation (SPM and XPM), which lower the achievable generalized signal-to-noise ratio (GSNR) and consequently shortens the transmission reach of WDM channels.

3. Network Model

To assess the network performance of the FXC networks, we implemented a network simulator including the degradation in quality-of-transmission (QoT) caused by each network element and fiber. The QoT metric is defined for the i-th channel as $\mathrm{GSNR}_i = P_{\mathrm{Rx}_i}/(P_{\mathrm{ASE}_i}+P_{\mathrm{NLI}_i}) = (\mathrm{OSNR}_i^{-1}+\mathrm{SNR}_{\mathrm{NL}_i}^{-1})^{-1}$, where P_{Rx_i} is the received signal power, P_{ASE_i} is the amplified spontaneous emission (ASE) noise power, and P_{NLI_i} is the nonlinear interference noise power, where $OSNR_i = P_{\mathrm{Rx_i}}/P_{\mathrm{ASE_i}}$ is the optical signal to noise ratio and $\mathrm{SNR}_{\mathrm{NL_i}} = P_{\mathrm{Rx_i}}/P_{\mathrm{NLI_i}}$ is the nonlinear interference (NLI) SNR, in the same bandwidth [2]. The simulation is performed for a C-band optical system with a total bandwidth of 4.8 THz. The per channel launch power is set to be 5 dBm. The nodes in the FXC network have a total loss of 12 dB. The amplifiers are based on commercially available Erbium-doped fiber amplifiers (EDFAs). The EDFA's are set at transparency, i.e., at the nominal gain, completely recovering fiber losses [8]. The noise figure is 5.5 dB with a flat gain ripple profile. Furthermore, standard SMF is used across all links and considers fiber-linked impairments, such as dispersion (D) = 16 ps/nm/km and attenuation (α) = 0.2 dB/km. The transmission system employs 800ZR+ transceivers with a channel spacing and bit rates reported in Table 1 [6]. To account for laser frequency inaccuracy and drift, we include a 1 GHz guard when computing the effective channel spacing, i.e., $\Delta f_{\mathrm{eff}} = \Delta f_{\mathrm{nominal}} - 1$ GHz. All reported $\Delta f/S_r$ ratios are therefore calculated using Δf_{eff}. The network evaluation is performed on the German (G50) and Japanese (J25) networks, illustrated in Fig. 1, where the red dots represent regional optical nodes while the green dots represent core nodes. Finally, the traffic in both networks follow a hierarchical distribution, where high-capacity flows are exchanged among core nodes, while regional nodes primarily connect to their two nearest core nodes. This results in two dominant classes of demand: core-to-core (C2C) traffic representing backbone exchanges, and core-to-regional (C2R) traffic. The model reflects realistic traffic asymmetry between major hubs and peripheral regions. The demand matrix for both the classes are expressed as number of channels, C2C traffic request is uniform randomly picked in range of 3 to 8 channels, while C2R is in evenly drawn in the range of 2 to 4 channels. The simulation uses 100 Monte Carlo iterations, and all results are averaged across these runs. All MNC values are reported at a 1% blocking probability. A request is blocked if the routing algorithm cannot find a transparent light path with enough spectral resources to allocate it.

4. Results and Discussion

To evaluate the combined impacts of ADXT and NLI under fiber switched operation, two realistic continental-scale topologies are analyzed: G50 and J25, both with 10 bidirectional pairs of fibers/link. Fig. 3a and 3b show the spectral grid spacing is progressively narrowed from 150 GHz to 120 GHz. Starting from a range of 32 to 35 channels/fiber (CpF) ($\Delta f/S_r \approx 1.14$ to 1.04, S_r=131.35 Gbaud), probabilistic constellation shaping (PCS) 16QAM-800G is feasible which provides roughly 1 dB shaping gain in required OSNR [6]. In this region, the J25 shows a smooth rise of ≈+9% MNC, while the G50 network reaches ≈+14% by 34 CpF. At 34 CpF, ≈ 18% of paths utilize PCS-16QAM 800G. When spacing tightens to 35 CpF, a slight GSNR reduction causes about (≈ 14%) of total PCS-16QAM 800G paths downgrade to 600G, leaving only 4% (18%-14%) of the 800G paths. The added wavelength compensates for this loss, yielding a narrow throughput plateau between 34 CpF and 35 CpF. The RRC=0.1 and 0.2 behave almost identically in this region. In CpF ranging 36 to 37 ($\Delta f/S_r \approx 1.12$ to 1.08, S_r=118.74 Gbaud), the PCS-16QAM 800G is not feasible due to the narrow channel bandwidth ($\Delta f/S_r \approx 1.05$, Δ f= 133.33 GHz S_r=131.35 Gbaud), and ADXT penalties also increase The J25 exhibits a clear dip for both RRCs



(a) **(b)**

Fig. 3: *Evolution of MNC at 1% blocking probability and percentage of MNC growth w.r.t number of channels/fiber assuming RRC 0.1 or 0.2 pulse shaping applied to 800ZR+ carriers, for (a) G50 and (b) J25.*

at 36 CpF. At 35 CpF, ≈15% of J25 paths are utilizing PCS-16QAM 800G, but as the spacing tightens to 36 CpF, all these paths transition to 600G. In addition, the longer average link length of J25 (318.70 km [3]) compared to G50 (193.14 km [3]) further lowers the GSNR margin, causing some 16QAM 800G paths on long routes to downgrade or become blocked, leading to the observed dip at 36 CpF. Beyond this point, both RRC cases exhibit a steady throughput recovery from 37 to 38 CpF. In contrast, the G50 network exhibits a steady increase even though PCS-16QAM 800G is no longer feasible. The few remaining PCS-16QAM 800G paths (4%) at 35 CpF compared to 15% in J25 shift to 600G. This fraction is too small to offset the added wavelength, resulting in a net ≈+15% MNC growth: RRC 0.2 rises steadily from 36 to 38 CpF, while RRC 0.1 extends this growth up to 39 CpF. In the 38 to 40 CpF range ($\Delta f/S_r \approx 1.06$ to 1.01, S_r=118.74 Gbaud), the combined impacts of ADXT and NLI become more dominant, defining the *critical compression region* where further spectral compaction no longer yields net efficiency gain. In the J25 network, the capacity trend peaks at 38 CpF, where ADXT coupling first becomes comparable to NLI. Both RRCs attain their maximum MNC percentage increase ≈+15%, before both curves begin to decline at tighter spacing. Beyond 39 CpF, ADXT dominates; RRC=0.1 retains a modest gain ≈+11%, whereas RRC=0.2 drops down to ≈+5.5% gain, confirming that the optimum compression point for longer average link length topologies lies near 38 channels. The earlier roll-off observed in J25 originates from its longer average link length, which reduce GSNR margins and increase sensitivity to ADXT under tighter channel spacing. In contrast, the G50 network, with shorter average link length, distributes nonlinear buildup more uniformly and maintains higher GSNR margins. Consequently, RRC=0.1 continues increasing up to 39 CpF, reaching its maximum increase in MNC ≈+21%, while RRC=0.2 begins to flatten earlier. The subsequent decline at 40 CpF results from compounded ADXT and accumulated NLI penalties. An irregularity in the MNC growth is observed as the channel spacing narrows. This non-monotonic behavior arises from the discrete nature of network routing and spectrum allocation, coupled with topology-dependent variations in GSNR across different paths. The denser topology of G50 indicates a possible trend, where short-span, high-connectivity networks might tolerate slightly tighter grids spacing before breakdown but tend to experience faster deterioration beyond the optimum. However, further analysis across additional topologies would be required to confirm this behavior.

5. Conclusion

This work shows the joint impact of ADXT and nonlinear effects on FXC architecture. The results show that the spectral grid tightens from 150 GHz to 120 GHz increases MNC by up to 15% and 21% for J25 and G50 networks, respectively. This MNC growth is less than the theoretical 25% channel-count gain, due to the combined ADXT and nonlinear penalties. These findings define practical limits for future FXC architecture based on 800ZR+ carriers.

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